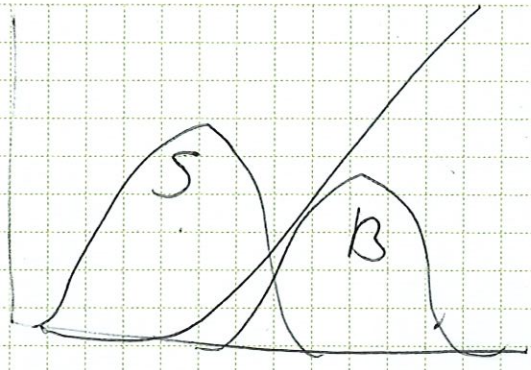


$$\frac{C}{1-C} = \frac{p(x|S)}{p(x|B)}$$

Samples, S/B → Density ratio

Samples x → Density of x → Sample

Generative networks

Image generators	Fast simulation,
LLM	Density estimation

$z \sim p_{latent}$

$z \sim \mathcal{N}(0,1)$

$G_\theta(z) \to x$ $\quad x \sim p_{data}$

$x \sim p_{model} \approx p_{data}$

Normalizing Flow

$$\hat{\theta} = \arg\max_\theta p(\theta \mid x_{train}) \propto \arg\max_\theta p(x_{train} \mid \theta)$$

$$\arg\max_\theta \prod_{x_i \in x_{train}} p(x_i \mid \theta)$$

$$\rightarrow L = \frac{1}{N} \sum_{x_i \in x_{train}} \log p_{model}(x_i \mid \theta)$$

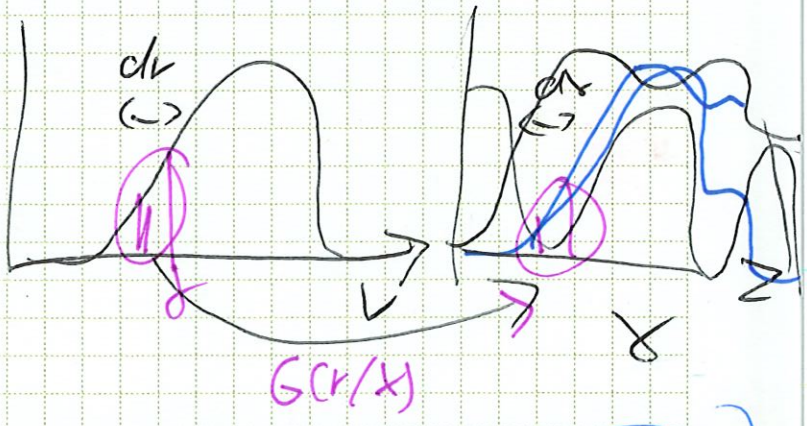
p_{model} ?

G: bijective map
probability conserving
continuous, monotonic

latent r $\xleftrightarrow[G_\theta^{-1}(x)]{G_\theta(r)}$ data x

$p_{latent} \rightarrow \mathcal{N}(0,1)$

$p_{model}(x)\,|dx| = p_{latent}(f)\,|dr|$



$z = G_\theta(x)$

$p_{model} = p_{data}(x) \left| \frac{dx}{dz} \right|$

$\Rightarrow p_{data}(G_\theta(x)) \left| \frac{dG_\theta(x)}{dx} \right|$

In D dimensions

$p_{model} = p_{data}(G_\theta(x)) \left| \det \frac{\partial G_\theta(x)}{\partial x} \right|$

$J_\theta(x) = \frac{\partial G_\theta(x)}{\partial (x)} \Rightarrow (J_\theta)_{ij} = \frac{\partial G_{\theta,i}(x)}{\partial x_j}$

$i,j : 1 \dots D$

$dz = dx \, |\det J_\theta(x)|$

$p_{model}(G_\theta(x)) \, |\det J_\theta(x)|^{-1} = p_{data}(z)$

Invertiba transformation

$$\begin{pmatrix} x_1 \\ \vdots \\ x_D \end{pmatrix} \longrightarrow \begin{pmatrix} r_1 \\ \vdots \\ r_D \end{pmatrix}$$

$$\vec{x} \cdot S + t = \vec{r} \qquad (\vec{r} - t) = \vec{x}$$

$$S(\vec{x}) + t(x) \qquad S(\vec{x})$$

$$\begin{pmatrix} x_1 \\ \vdots \\ x_{\frac{D}{2}-1} \end{pmatrix} \vec{x}' \quad \vec{r} = \vec{x} \cdot S + t \quad S(\vec{x}') + t(\vec{x}') \begin{pmatrix} r_1 \\ \vdots \\ r_{\frac{D}{2}-1} \end{pmatrix}$$

$$\begin{pmatrix} x_{\frac{D}{2}} \\ \vdots \\ x_D \end{pmatrix} \vec{x}' \longrightarrow \begin{pmatrix} r_{\frac{D}{2}} \\ \vdots \\ r_D \end{pmatrix}$$

$$\vec{r}' = \vec{x}'$$

$$S(\vec{x}') = S(\vec{r}')$$

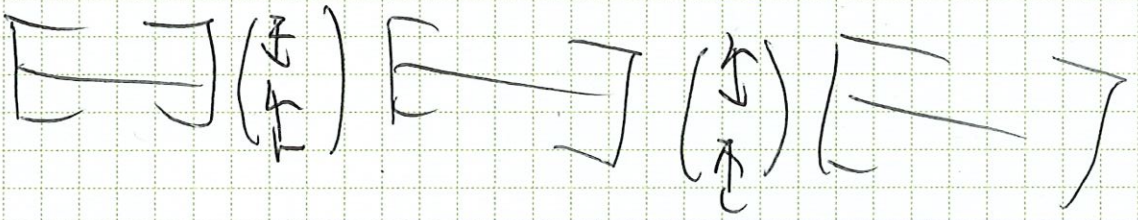
$$\vec{x}' = \frac{\vec{r} - t(\vec{r}')}{S(\vec{r})}$$

Invertiba Model

$$\vec{r} = \vec{x} e^{S(\vec{x})} + \vec{c}$$

$$J_G \left(\begin{matrix} \text{diag}(\exp S(\vec{x})) & \frac{\partial x'}{\partial r^\mu} \\ 0 & \mathbb{1} \end{matrix} \right) =$$

$$\det J_G \prod \exp S(\vec{r}^{\,a})$$



$$G_G(k) = x \qquad G_G(x) = k$$

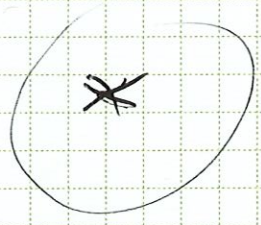
~~$\operatorname{argmax}_\Theta \left(p_{model}(x|\Theta) \right)$~~

~~$\operatorname{argmax}_\Theta$~~

$$L = \frac{1}{N} \sum_{x_i \in x_{train}} \log p_{model}(x|\Theta)$$

$$\| \cdot p_{calc}\left(G_\Theta^{-1}(x)\right) \left| \det J_{G^{-1}}(x) \right| ^{-1}$$

$$\log p_{Gauss}\left(G_\Theta^{-1}(x)\right) + \log \det J_{G^{-1}}(x)$$

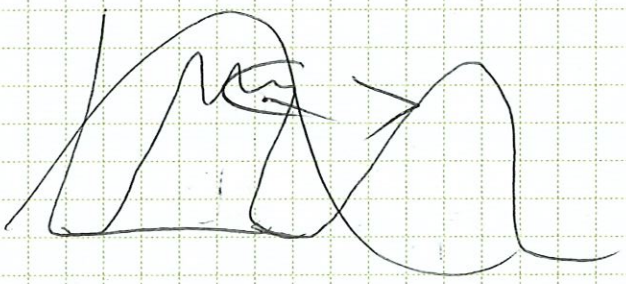
$L \sim$ Gauss $\quad G(x) \Rightarrow$ 



diffusion models

Data -> Gaussian -> add noise

data x_0 ⟷ Gaussian (add noise / denoise)



$x_t|_{t=0}$: data

$$x_t = \sqrt{1-\beta_t}\, x_{(t-1)} + \sqrt{\beta}\,\varepsilon \qquad \varepsilon \sim \mathcal{N}(0,1)$$

$$x_1 = x_0$$

$\beta_0 \dots \beta_T$, $t: 0 \dots T$ noise schedule

-> optimize like NN

$$\alpha_t = 1-\beta_T = \bar{\alpha}_t = \prod_{s=0}^{t} \alpha_s$$

$$x_t = \bar{\alpha}_t x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon$$

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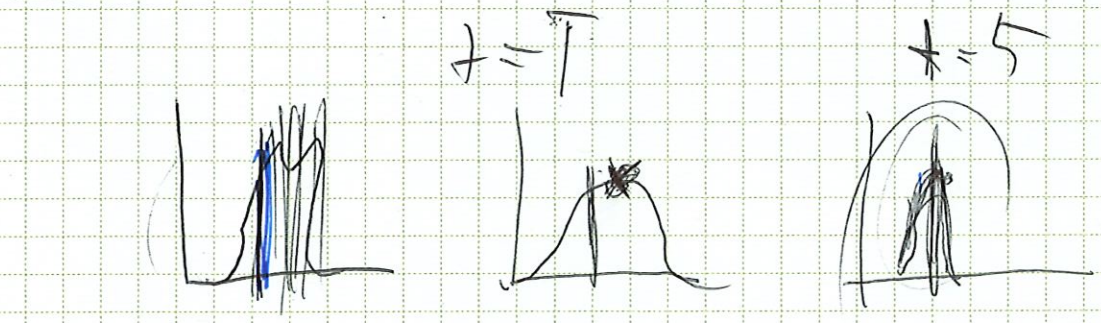
$$x_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon}{\sqrt{\bar{\alpha}_t}}$$

$$\epsilon_\theta(x_t, t) \approx \epsilon$$

L_{DDPM} (Denoising Diffusion Probabilistic model)

$$= \left(\epsilon_\theta(x_t, t) - \epsilon\right)^2$$

x_0 ~~$t = T$~~ $t = T$



Generate:

$$x_T \sim \mathcal{N}(0,1)$$

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_T, t)\right) + \sqrt{\beta_t}z$$

$z \sim \mathcal{N}(0,1)$

$\rightarrow x_0 \rightarrow$ new ~~image~~ data

Conditional flow matching

$$x_t = (1-t)x_0 + t\epsilon, \quad \epsilon \sim \mathcal{N}(0,1)$$

$$\frac{dx_t}{dt} = -x_0 + \epsilon = (\epsilon - x_0)$$

$$\rightarrow L_{CFM} = \left(N_\theta(x_t, t) - (\epsilon - x_0)\right|^2 \quad x_0 = \text{data}$$

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$$x_{t-\Delta t} = x_t - \Delta_t \frac{dx_t}{dt} = x_t - \Delta t \cdot v_\theta(x_t, t)$$

→ Edg Solving diffeq. $\frac{dx_t}{dt} = v_\theta(x_t, t)$

Conditional Densities, Conditional Gen. Model

$x_{data} \sim p_{data}$ $p_{model} \approx p_{data}$

Image (prompt)

$x \sim p_{Images}(x|C)$

↑ prompt

$p_{model}(x|C, \Theta)$